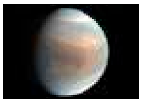
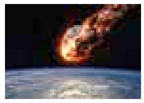
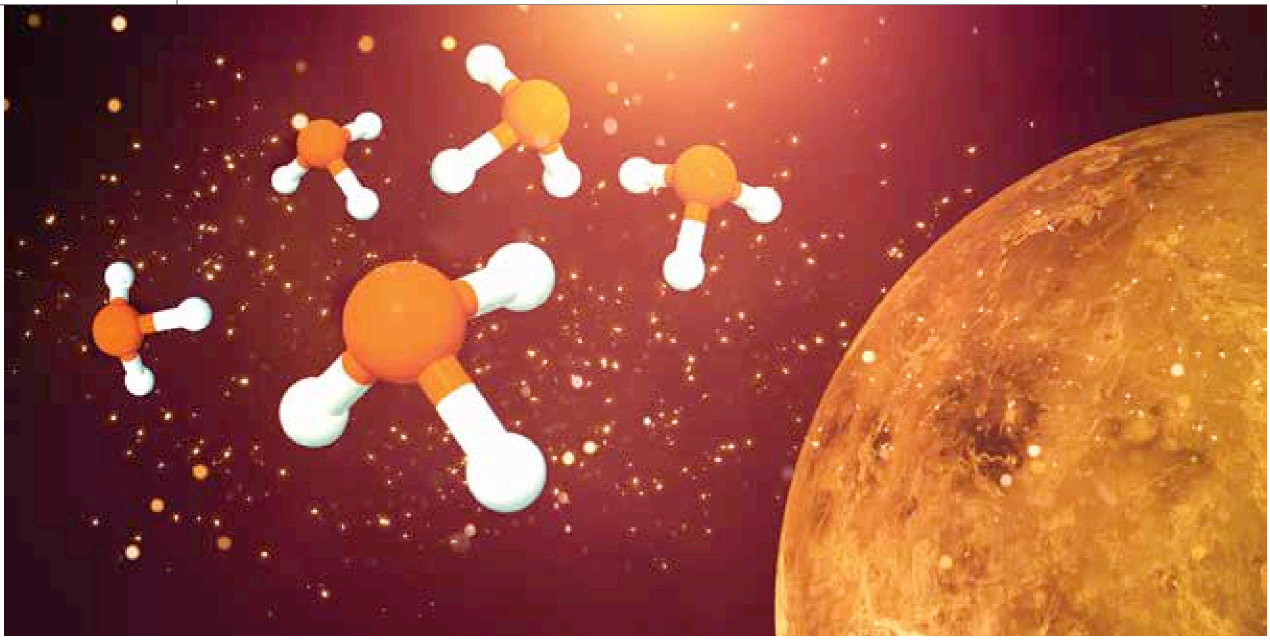


Race to find life on Venus
US-based aerospace company, Rocket Lab looks to beat NASA in search of life on Venus. **http://dgit.in/oct20-43**



Meteors brought life to Venus?
Research suggests asteroids that grazed Earth's atmosphere brought microbes to Venus from Earth. **http://dgit.in/oct20-44**



LIFE ON VENUS

Astronomers have discovered a biosignature next door

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Venus has a thick, hot atmosphere that is not considered suitable for life, at least as we know it on Earth. Millions of years ago, it was considered to be a desert planet, similar to how Mars is at this point in time. It was also believed that further back in time, about two billion years ago, Venus could also have liquid water on the surface, where life would emerge. Over the years, the buildup of atmospheric gases, primarily through volcanism, gave Venus its dense atmosphere. This atmosphere is made up mostly of carbon dioxide (95 percent). The other important gases that make up the chemical composition of the Venusian atmosphere are nitrogen and sulfur dioxide. Both carbon dioxide and sulfur dioxide are greenhouse gases, that trap heat in the atmosphere, as a consequence of which, Venus has a surface temperature of around 467 °C, which makes it the hottest planet in the Solar System. The atmosphere is also extremely dense, with atmospheric pressures reaching 93 times that of the Earth at the surface. Because of the bright sulfur dioxide clouds suspended in the atmosphere, Venus is also the brightest planet in the Solar System. The clouds are so thick that they make it difficult to observe the surface. Apart from the heat and pressure, the atmosphere on Venus is also very dry. There is less water vapour in the atmosphere than the driest deserts on Earth.

At an altitude of about 50 kilometers though, the atmospheric conditions of pressure and temperature are similar to the surface on Earth. The temperature in the atmosphere is a relatively cool 60°C. At this particular altitude, Venus is the most Earth-like planet we know of. If at some point in the past (say 2 billion years ago), if life did emerge on the surface of Venus, then it would have to adapt and move, forced to the narrow band in the clouds where it can thrive, far away from the heat, dryness, pressure and volcanism. Scientists have long known of this particular region in the Venusian atmosphere that was conducive to life. Now, astronomers have found a biomolecule, phosphine, at this very altitude. While biomolecules have been identified on Venus before, they could also be produced by other geological or chemical processes. What makes the identification of phosphine different is that there are simply no know geological or chemical processes on terrestrial planets that can produce the gas. As far as we know, there is only one thing that can produce phosphine, and that thing is life.

Researchers from MIT had previously shown that if phosphine were to be discovered on a terrestrial planet, it could only come from life. While researchers had identified phosphene as a promising biomarker to identify exoplanets with life, they had not really looked at planets within the Solar System. Now, astronomers from Cardiff University have discovered phosphine on Venus. Phosphine is a colourless, toxic and flammable gas that smells like rotting fish. The discovery of